

## Solution

### GRAVITATION-01

#### Class 09 - Science

1.

(d) 0.83

**Explanation:** According to Archimedes's principle, buoyant force,  $F_B$  = Volume of the object immersed in liquid  $\times$  density of liquid  $\times$  acceleration due to gravity. Let  $V$  be the volume of object.

$$\text{For water: } F_B = \frac{3V}{4} \times \rho_w \times g = \frac{15}{2} V \rho_w \dots (i)$$

$$\text{For alcohol: } F_B = \frac{9V}{10} \times \rho_a \times g = 9 V \rho_a \dots (ii)$$

From eqn. (i) and (ii),

$$\frac{15}{2} V \rho_w = 9 V \rho_a \therefore \frac{\rho_a}{\rho_w} = \frac{15}{2 \times 9} = \frac{15}{18} = 0.83$$

2.

(b) universal gravitational constant

**Explanation:** We know that the gravitational force  $F = G \frac{Mm}{r^2}$

Where  $F$  = force between mass

$m$  = mass of the body

$G$  = gravitational constant

$r$  = Distant between two mass

Given  $M = 1$  unit and  $r = 1$  unit, we get

$$F = G \frac{1 \times 1}{1^2}$$

$$\Rightarrow F = G$$

3.

(c) revolution of the earth around the sun

**Explanation:** The revolution of the earth around the sun takes place in a trajectory orbit. The average distance between the earth and the sun is approximately 150 million km.

4.

(b)  $20 \text{ ms}^{-1}$

$$\text{Explanation: } V^2 - u^2 = 2gs$$

$$u = 0 \text{ g } g = 10 \text{ ms}^{-1}, s = 20 \text{ m}$$

$$V^2 = 2 \times 10 \times 20 = 400$$

$$V = 20 \text{ m/s}$$

5.

(d) both density of the liquid and dipped volume of the body immersed.

**Explanation: Factors affecting the Upthrust are:**

i. Volume of object immersed in liquid - as volume increases the upward force increases.

ii. Density of liquid: As the density of liquid increases the upward force exerted also increases.

6.

(b) pressure

$$\text{Explanation: pressure} = \frac{\text{force}}{\text{area}}$$

So, if the area is more pressure exerted will be less and vice versa. This is the reason why the shoulder strap in school bags are broad. This helps in reducing the pressure exerted on the shoulders of the person and thus makes it more comfortable to carry around.

7.

(a) thrust

**Explanation:** Force exerted by an object perpendicular to the surface is called thrust.

8.

(b) R

**Explanation:** Weight of passenger on Earth =  $m \times g_e = 60 \times 10 = 600 \text{ N}$

Weight of passenger on Mars =  $m \times g_m = 60 \times 4 = 240 \text{ N}$

As the passenger is going from the Earth to the Mars in a spaceship moving with a constant velocity so the acceleration due to gravity because of Earth and Mars becomes negligible (zero) at some point in space. Thus, weight of passenger becomes zero. After sometime, as he reaches near to the Mars, his weight increases because of acceleration due to gravity of Mars. Finally, at the surface of Mars, his weight becomes 240 N.

9.

(c) 78.4 m

**Explanation:**  $S = ut + \frac{1}{2}gt^2$  where S = distance, U = initial velocity, t = time and g = acceleration due to gravity.

Given: U = 0 m/sec, t = 4sec.

we know  $g = 9.8 \text{ m/sec}^2$

Therefore,  $S = \frac{1}{2}gt^2$

$S = \frac{1}{2} \times 9.8 \times 16$

S = 78.4 meter.

10.

(c) (a) - (i), (b) - (iii), (c) - (iv), (d) - (ii)

**Explanation:**

| Column I                 | Column II                             |
|--------------------------|---------------------------------------|
| (a) g at height h        | (i) $g \left(1 - \frac{2h}{R}\right)$ |
| (b) g at depth h         | (ii) $g \left(1 - \frac{h}{R}\right)$ |
| (c) $g_{\text{poles}}$   | (iii) Maximum                         |
| (d) $g_{\text{surface}}$ | (iv) $\frac{GM}{R^2}$                 |

11.

(b)  $1.6 \times 10^4 \text{ N m}^{-2}$

**Explanation:** Density of oil ( $\rho_o$ ) = 0.60 g/cc = 600 kg/m<sup>3</sup>

Density of water ( $\rho_w$ ) = 1 g/cm<sup>3</sup> = 1000 kg/m<sup>3</sup>

Pressure due to water =  $h\rho_w g = 1 \times 1000 \times 10 = 10000 \text{ N m}^{-2}$

Pressure due to oil =  $h\rho_o g = 1 \times 600 \times 10 = 6000 \text{ N m}^{-2}$

Total pressure = 10000 + 6000 = 16000 N m<sup>-2</sup> =  $1.6 \times 10^4 \text{ N m}^{-2}$

12.

(d) varying velocity

**Explanation:** The Earth's orbit is not perfectly circular. As a result its velocity changes slightly over the course of the year.

13.

(d) Greater on earth and lesser on Moon

**Explanation:** Weight depends on gravity,  $W = mg$

As the gravity of Earth is greater than the gravity of the moon, the object weighs more on Earth.

14.

(d) least in salt solution

**Explanation:** The buoyant force acting on the solid depends on the factors such as the density of fluid and volume of solid. As the density of saltwater is highest among the given options, the buoyant force is maximum in the case of saltwater and the reading of spring balance would be least in this case.

15.

(a) all of these

**Explanation:** The amount of heat energy lost by the hot body in a given time depends on temperature, surface nature and the area affect the loss.

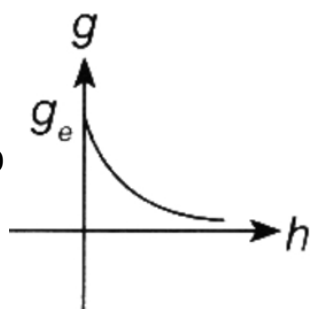
16. **(d)** Increases  
**Explanation:** At Antarctica, the value of acceleration due to gravity  $g$  is more, therefore, the weight of sugar bag will increase. We know at the equator, acceleration due to gravity is minimum and at the pole (Antarctica is at the South Pole), it is maximum. Therefore, the weight of the bag increases in Antarctica.
17. **(d)** gravity  
**Explanation:** Earth's atmosphere is the layer of gases around the Earth. The atmosphere is held in place by Earth's gravity.
18. **(d)** zero as long as no other mass is inside.  
**Explanation:** The gravitational force on an object of mass placed inside a spherical shell of matter is zero.
19. **(c)**  $2R$   
**Explanation:** We know,  
 $g = \frac{GM}{R^2}$  ... (i)  
 Also  $g' = \frac{GM}{(R+h)^2}$  ... (ii)  
 Equating eqns. (i) and (ii), we get  
 $9R^2 = (R + H)^2$   
 or  $3R = R + h$  or  $h = 3R - R = 2R$
20. **(b)** All of these  
**Explanation:** Acceleration due to gravity varies with height, depth and shape of the planet as follows-
  - As the height (altitude) increases acceleration due to gravity ( $g$ ) decreases.
  - As the depth increases acceleration due to gravity ( $g$ ) decreases.
  - Acceleration due to gravity is inversely proportional to the radius.
21. **(c)**  $p = q$   
**Explanation:** Gravitational force is same in both the conditions because gravitational force between two objects is independent of the surrounding medium.
22. **(d)** Dependent on mass of the planet for points outside the planet.  
**Explanation:** The gravitational force exerted by any planet is based on the effective mass of the planet, which is in turn dependent on if you are on the surface of the planet, above the surface or below the surface of the planet.  

$$F = \frac{Gm_1m_2}{r^2}$$
23. **(a)** weight of water displaced by the solid  
**Explanation:** Weight loss = upthrust. So, app. weight =  $W_{\text{air}} - W_{\text{lost}}$
24. **(c)**  $2\text{ N}$   
**Explanation:** Weights of air =  $10\text{ N}$   
 When immersed fully in water =  $8\text{ N}$   
 The weight displaced =  $10\text{ N} - 8\text{ N} = 2\text{ N}$
25. **(d)**  $W_1 = W_2$   
**Explanation:** As the spring balance is kept in a satellite, it experiences weightlessness.  
 So,  $W_1 = W_2 = 0$   
 Even if satellite changes its orbit, weight will not change.

26.

(c) at poles

**Explanation:**  $F = G (mM/R^2)$ , acts along the line joining the centre of the earth and the centre of the body. This force produces an acceleration 'g' in the body of mass m. Also, Acceleration due to gravity is proportional to the mean density of the earth, and its radius.



27. (a)

**Explanation:** Value of g decreases with height.

28.

(d) 4 times

**Explanation:** The force of gravitational attraction between the two bodies of mass (M) and (m) is

$$F = G \frac{Mm}{r^2}$$

Now, If masses of both objects are halved without changing distance between them or  $M = 2M$  and  $2m$

$$F = G \frac{2M \times 2m}{r^2} = 4G \frac{Mm}{r^2}$$

Thus, force becomes four times as we increase the masses to double without altering the distances.

29.

(b) zero

**Explanation:** The weight of the object is zero

Because weight = mg

And  $r = 0$  so  $g = \text{zero}$  at the center of earth

Then  $w = \text{zero}$

30.

(d) none of these

**Explanation:** The gravitational constant is independent of mass, distance, and temperature.

31. (a) Newton

**Explanation:** The force of attraction of Earth on a body is the same as the weight of the body.

Therefore, the Weight of an object is measured in the same unit as Force.

32.

(d) will move along a straight line tangential to the circular path

**Explanation:** In circular motion, the direction of velocity at a point is always along the tangent at that point. If string breaks, then the centripetal force acting on the stone becomes zero and it will move along a straight line tangential to the circular path.

33.

(d) it is completely immersed in the liquid.

**Explanation:** When a solid is completely immersed in the liquid, there is a maximum apparent loss in its weight due to the maximum volume of liquid displaced.

34. (a) 100 N

**Explanation:** Weight =  $Mg = 10 \times 10 = 100 \text{ N}$

35. (a) increases till it attains the weight in air

**Explanation:** Increases till it attains the weight in air as-

The buoyant force exerted by a fluid exerts upwards on a body submerged in it,

$$F = \rho V$$

$$F = \rho Vg$$

where  $\rho$  is the density of the fluid, V is the volume of the fluid displaced, and 'g' is the acceleration due to gravity.

The liquid pushes the solid upward and by Newton's third law, there is a force of equal magnitude acting downward on the liquid.

36. (c) maximum when breadth and width from the base  
**Explanation:** Now, according to question, when base is formed by breadth and width. Area will be minimum. And so, pressure will be maximum.
37. (c)  $2.45 \text{ ms}^{-2}$   
**Explanation:** g is inversely proportional to the square of the distance between the earth and the body. As the distance gets twice the g will become one - fourth.
38. (d) Weight  
**Explanation:** Loss of weight is equal to the upthrust.
39. (d)  $g \propto \frac{1}{r^2}$   
**Explanation:**  $g = \frac{GM}{r^2}$   
 So,  $g \propto M$  and  $g \propto \frac{1}{r^2}$
40. (d) the gases have less requirement of velocity or energy to escape from its surface.  
**Explanation:** Escape velocity is the speed that an object needs to be traveling to break free of a planet or moon's gravity well and leave it without further propulsion.
41. (a) 10 kg  
**Explanation:** Gravity on the Moon has approximately 1/6th of the strength of gravity on Earth, a man who weighs 60kg on Earth would weigh approximately 10kg on the Moon. His mass, however, remains constant.
42. (d) increase  
**Explanation:** The value of 'g' will slightly increase because if the earth stops rotating then it will be more spherical.
43. (a)  $\frac{24}{25} h \text{ m}$   
**Explanation:** Here, height = h m, time =  $\frac{T}{5} \text{ s}$   
 As  $s = ut + \frac{1}{2}at^2 \therefore h' = \frac{1}{2}g\left(\frac{T}{5}\right)^2$  or  $h' = \frac{1}{25}h$   
 Position of the ball above the ground  
 $= h - h' = h - \frac{h}{25} = \frac{24h}{25} \text{ m}$
44. (a) Statement (iv) is correct.  
**Explanation:** The acceleration due to gravity is always directed downwards towards the centre of the Earth for a freely falling body.
45. (b) More in sea water.  
**Explanation:** Loss in weight depends on upthrust which is dependent on the density of the liquid besides the volume immersed. Upthrust is more in seawater and so the loss in weight is more in it.
46. (a) Difference in density of the body and half the density of the liquid.  
**Explanation:** Weight = W - U  
 $V\rho g - \frac{V}{2}\sigma g = Vg\left(\rho - \frac{\sigma}{2}\right)$
47. (a) any two bodies having some mass  
**Explanation:** Law of gravitation is applicable to all bodies having some mass. And is given by  $F = \frac{Gm_1m_2}{r^2}$   
 where, F = force of attraction between the two bodies  
 $m_1, m_2$  = mass of two bodies  
 G = Gravitational constant  
 r = Distance between the two bodies.

48. (c) equally  
**Explanation:** Acceleration =  $\frac{\text{change in velocity}}{\text{time taken}}$   
 Thus the change in velocity with respect to the time taken will be equal in each of the respective cases.
49. (a) heavier than water and insoluble in it  
**Explanation:** It should be heavier than water and insoluble in it, so that it sinks and displaces the water and also it does not mix in the water.
50. (b) 9.8  
**Explanation:**  $W = m \times g = m \times 9.8$   
 $\frac{W}{m} = 9.8$
51. (a) Buoyant force  
**Explanation:** The upward force experienced by the body when submerged in a fluid is called buoyant force. Pressure increases with depth as a result of the weight of the overlying fluid. So, the pressure at the bottom of an object submerged in a fluid is greater at the top of the object. This pressure difference results in a net upwards force on the object. Buoyant force depends upon nature of fluid and density of substance.
52. (b)  $W_A > W_B$   
**Explanation:** As density of salty water is more than the density of tap water. It will exert a greater buoyant force (upthrust) and, So,  $W_A$  is greater than  $W_B$ .
53. (c)  $P_A = P_B$   
**Explanation:** Pressure varies with height of liquid in a container and height is same for both the points so pressure at both points remain same.
54. (c) 2.17 N  
**Explanation:** Here,  $M_p = 2 M_e$  and  $R_p = 3R_e$   
 $m = 1 \text{ kg}$ ,  $W_p = ?$   
 As  $g_e = \frac{GM_e}{R_e^2}$  and  $g_p = \frac{GM_p}{R_p^2}$   
 $\frac{g_p}{g_e} = \frac{M_p}{R_p^2} \times \frac{R_e^2}{M_e}$   
 or  $g_p = g_e \times \frac{M_p}{M_e} \times \left(\frac{R_e}{R_p}\right)^2$   
 or  $g_p = 9.8 \times 2 \times \left(\frac{1}{3}\right)^2 = 2.17 \text{ m/s}^2$   
 $W_p = mg_p = 1 \times 2.17 = 2.17 \text{ N}$
55. (a) Less  
**Explanation:** Pressure is inversely proportional to cross-sectional area.
56. (b) lighter will have more K.E.  
**Explanation:** The lighter body will have more K.E because kinetic energy is inversely proportional to the mass of the body.
57. (a) is independent of mass and radius of the earth  
**Explanation:** The value of  $g$  is independent of the mass of the object and only dependent upon location - the planet the object is on and the distance from the center of that planet.
58. (a) = 0  
**Explanation:** When a body floats in a liquid, upthrust acting on the body is equal to the weight of the body in accordance with the law of flotation. That is, the weight of the body is balanced by the upthrust. So, the apparent weight of the body is zero as the net force acting on the body is zero.

59. (b) have same velocities at any instant  
**Explanation:** Object of different masses falling freely near the surface of the moon would have the same velocities at any instant because they will have same acceleration due to gravity.
60. (d) 44%  
**Explanation:** Momentum  $mv$  increase by 20% if velocity  $v$  increase by 1.2  $v$ .  
 Now, kinetic energy  $k = \frac{1}{2} mv^2$ , increase  $KE = \frac{1}{2} m(1.2v)^2 - \frac{1}{2} mv^2 = \frac{1}{2} mv^2 (1.44 - 1) = 0.44 \times \frac{1}{2} mv^2 = 0.44 K$   
 $\therefore$  Percentage increases in  $KE = \frac{0.44k}{k} \times 100 = 44\%$
61. (c)  $U = W$   
**Explanation:** Weight is a force acting on an object and always acts downwards. Upthrust is a force exerted by fluid in an upward direction. For an object to float both of these forces act simultaneously and eventually get balanced with each other.
62. (b) In the direction opposite to the direction of motion.  
**Explanation:** When a ball is thrown then the direction of motion of the ball and vertically upwards then the direction of motion velocity is the same, i.e. upward of the ball is upwards but the acceleration due to gravity acts downwards towards the earth. Thus, the direction of acceleration is opposite to the direction of motion of the ball.
63. (c) is universal constant of nature  
**Explanation:** Law of gravitation is applicable to all bodies having some mass. And is given by  $F = \frac{Gmm}{d^2}$   
 where,  $F$  = force of attraction between the two bodies  
 $m$  = mass of two bodies  
 $G$  = Gravitational constant  
 $r$  = Distance between the two bodies.
64. (b) All of these  
**Explanation:** Density, mass and radius of moon is different than earth so the acceleration due to gravity is different and calculatively obtained 1/6th of Earth's gravity.
65. (d) 1 m  
**Explanation:** Relative positions remain constant irrespective of the distance travelled and time taken during free fall.
66. (c)  $U = W_a - W_l$   
**Explanation:** Up thrust = apparent lost in weight of the object  
 Up thrust =  $W_a - W_l$
67. (b)  $\frac{10^6}{5}$   
**Explanation:** Area =  $\pi r^2 = 0.50 \text{ mm}^2 = 0.5 \times 10^{-6} \text{ m}^2$   
 Let  $L$  be its length.  
 So volume,  $V = 0.5 \times 10^{-6} L \text{ m}^3$   
 Weight in air,  $W_a = 75 \text{ N}$   
 Weight in water,  $W_w = 65 \text{ N}$   
 So buoyant force by water =  $W_a - W_w = 10 \text{ N}$   
 $= V \times \text{density of water} \times g$   
 or  $10 = 0.5 \times 10^{-6} L \times 1000 \times 10$   
 $L = \frac{10}{0.5 \times 10^{-2}} = \frac{10 \times 10^2}{0.5} = \frac{10^4}{5} \text{ m} = \frac{10^6}{5} \text{ cm}$

68. (c) Negative energy means it is bound  
**Explanation:** Negative energy means it is bound and the work done by the gravity is the positive inbound state when it is approached by some mass. The mass gets trapped under the potential energy of gravity and is not allowed to escape.
69. (c) Because of greater mass of earth  
**Explanation:** Acceleration is inversely proportional to mass according to Newton's 3rd law of motion. So, a small stone produces large acceleration. Due to the large acceleration of stone, it falls towards Earth.
70. (c) Zero  
**Explanation:** The upthrust and the weight will be equal.
71. (b) 2.5 g wt  
**Explanation:** L.C. =  $\frac{25-0}{10} = \frac{25}{10} = 2.5 \text{ g wt}$
72. (b) is least on equator  
**Explanation:** the value of g on earth is maximum at poles and decreases as we go from poles to the equator and is minimum at the equator.
73. (b) any body thrown vertically up against gravity.  
**Explanation:** When an object is thrown vertically upwards at the maximum height the velocity is going to be zero, however, acceleration is not zero. If there is no velocity, there can not be acceleration.
74. (a) Both statement 1 and statement 2 are true but statement 2 is not the correct explanation of statement 1.  
**Explanation:** Variation of g with depth from surface of earth is given by  $g' = g \left(1 - \frac{d}{R}\right)$   
 At the centre of earth,  $d = R$ ,  $\therefore g' = g \left(1 - \frac{R}{R}\right) = 0$   
 $\therefore$  Apparent weight of body =  $mg' = 0$
75. (b) Statement 1 is false but statement 2 is true.  
**Explanation:** Inside the satellite,  $g = 0$   
 $T = 2\pi\sqrt{\frac{I}{g}} = 2\pi\sqrt{\frac{1}{0}} = \infty$   
 Thus, the pendulum will not oscillate.